

PES UNIVERSITY, Bangalore
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Department of Computer Science & Engineering
UE24CS341A : Software Engineering

LAB - 01

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Problem Statement - 64: Disaster Relief Supply & Volunteer Coordinator

Requirements Table

Actors: Relief Volunteer, Disaster Manager, Warehouse System

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match incoming field shelter supply requisitions against available central warehouse supplies and allocate delivery priority.	High	Pass: Critical needs (water/medical) are given the highest dispatch priority and added to the logistics manifest. Fail: A low-priority item is dispatched before a critical need.	To make sure critical supplies are delivered first.
FR-002	Functional	The system shall allow a Relief Volunteer to register and update their skill profile (e.g., medical, driving, first-aid) so that managers can search for the right person for a task.	Medium	Pass: When a volunteer saves a new/updated skill, it appears in the Disaster Manager's search results within 5 seconds. Fail: An updated skill is not reflected in search results.	To make sure users can access the system when it is needed.
FR-003	Functional	The system shall let a Disaster Manager dispatch a Relief Volunteer to a shelter by matching the task's required skill against volunteer skill profiles.	High	Pass: Only volunteers whose profile contains the required skill are shown as dispatch candidates. Fail: A volunteer without the required skill is dispatched to a task.	To assign volunteers to work they are suitable for.
FR-004	Functional	The system shall track the real-time location and delivery status (Pending / In-Transit / Delivered) of every relief vehicle assigned to a requisition.	High	Pass: Status updates automatically as GPS coordinates change and shows 'Delivered' once the vehicle enters the shelter's geofence. Fail: Status remains 'Pending' after delivery is confirmed on-site.	To make sure supplies can be sent to the required shelters.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall automatically alert the Disaster Manager if a dispatched vehicle does not reach its destination shelter within the estimated delivery time (ETA).	Medium	Pass: An alert is generated within 2 minutes of the ETA being exceeded. Fail: No alert is sent even though the ETA has been exceeded by 30+ minutes.	To help the manager monitor relief deliveries.
NFR-001	Non-Functional Performance & Security	The disaster mapping dashboard must operate with low network bandwidth consumption and support offline GIS data caching.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Dashboard fails to load cached map data when the connection drops.	To allow the system to work in disaster areas with poor network connectivity.
NFR-002	Non-Functional Reliability & Availability	The system shall be available to Disaster Managers and Relief Volunteers during relief operations.	High	Pass: Data entered while offline is queued locally and syncs automatically with no data loss once connectivity returns. Fail: Any offline entry is lost after the device reconnects.	To make sure users can access the system when it is needed.